

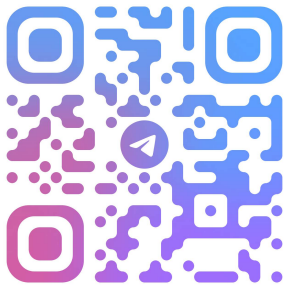
University of Information Technology & Sciences (UITS)
Faculty of Science and Engineering
Department of Computer Science & Engineering
Program: B.Sc. in CSE
Mid Term Examination, Autumn 2025
Course Title: Coordinate Geometry, Linear Algebra and Vector Analysis
Course Code: MATH 0541211

Marks: 20

Time: 1(one) hour

(Answer all questions)

- | Q.No. | Questions | Marks |
|-------|---|-------|
| 1. | a) Evaluate the lines represented by the given equation: $3x^2 - 16xy + 5y^2 = 0$. | [02] |
| | b) Find the value of k so that the following equation may represent pairs of straight line: $6x^2 + xy + ky^2 - 11x + 43y - 35 = 0$. | [02] |
| | c) The equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is transformed to the equation $4x^2 + 2y^2 = 1$ that referred to the rectangular axes through the point $(2,3)$ and inclined at an angle 45° . | [06] |
| 2. | a) Evaluate the angles and point of intersection between the lines represented by:
i) $x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$
ii) $34x^2 + 24xy + 41y^2 + 48x + 14y - 108 = 0$ | [04] |
| | b) Determine the nature of the equation and reduce to the standard form:
$5x^2 - 2xy + 5y^2 - 8x - 8y - 8 = 0$. | [06] |



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$$\frac{25}{2} + 3 = \frac{25+6}{2} = \frac{31}{2}$$